

Human-Centered Artificial Intelligence, Creativity, and Agency

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Introduction

In this inaugural volume of *Cadernos Estação*, focused on the intersections of technology, human agency, writing, and critical thinking, we are deeply honored to feature an exclusive written interview with Professor Ben Shneiderman, a global pioneer in Human-Computer Interaction (HCI) and a cornerstone reference in human-centered technology studies.

Prof. Shneiderman is a Distinguished University Professor Emeritus in the Department of Computer Science and a founding member of the Human-Computer Interaction Laboratory (HCIL) at the University of Maryland, as well as the author of the seminal book *Human-Centered AI* (Oxford University Press). His theoretical framework strongly advocates that artificial intelligence systems should amplify, augment, empower, and enhance human agency, rather than replace or diminish it.

Below, Prof. Shneiderman addresses crucial questions regarding the current culture of AI anthropomorphization, control-based interface design (supertools vs. autonomous teammates), reliable governance, and the future of human creativity in the

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² Distinguished University Professor Emeritus in the Department of Computer Science, a founding member of the Human-Computer Interaction Laboratory (HCIL), and a member of the Institute for Advanced Computer Studies (UMIACS) at the University of Maryland. He is globally renowned for his pioneering contributions to the field of Human-Computer Interaction (HCI), including the development of direct manipulation concepts (such as scrollbars and hyperlinks). He is the author of foundational works in the field, most notably *Human-Centered AI* (Oxford University Press, 2022).

era of large generative models. We warmly invite our readers to explore his foundational work further through his book and the publisher's official platform.

Editorial note:

The author's reference book is available on the Oxford University Press website at <https://global.oup.com/academic/product/human-centered-ai-9780192845290?cc=br&lang=en>

The Interview

1. On Supertools vs. Autonomous Teammates

Antonio Paulo: *In Human-Centered AI, you advocate for viewing AI systems as powerful "supertools" with high levels of human control and automation, rather than autonomous agents or pseudo-human teammates. In the current boom of Generative AI, how do you evaluate the cultural push to anthropomorphize these models, and what are the primary risks of treating software as a collaborator rather than a tool?*

Prof. Ben Shneiderman: *It's not only about "viewing AI systems as powerful", but it's about redesigning the current anthropomorphic designs that depend on a simple prompt window, when they could include richer features that remind users of possibilities. This approach, with check boxes, sliders, and form fill-in, helps users clarify their intent so as to get better control over the outcome. The goal of design should be to build user self-efficacy, support creativity, clarify responsibility, and facilitate social connections. I prefer to describe machines as tools or supertools, suggesting human control and responsibility. Supertools can provide a high degree of automation, but humans remain in control of essential features of a task.*

2. On Creativity and Authenticity

Antonio Paulo: *As generative language and image models become ubiquitous, there is a growing tendency toward stylistic homogenization and algorithmic predictability. How can the HCAI paradigm guide designers and educators to ensure that AI amplifies genuine human self-expression and exploratory creativity rather than flattening it into probabilistic averages?*

Prof. Ben Shneiderman: *By giving users greater control through improved control panels, users will be able to get what they want more often, while clarifying that they are responsible for the outcome. The improved user interfaces open the door to more creative explorations that go beyond what the AI system extracted from the training data and offer a greater potential for genuinely novel literary, artistic, musical, scientific, and other innovations. I called these systems tools or supertools, suggesting that the users are in control and responsible for the results.*

3. On Governance and Reliable Systems

Antonio Paulo: *A critical pillar of your book is the Two-Tier model of governance (organizational software engineering practices combined with independent external oversight and certification). Given the speed at which generative systems are being deployed in sensitive fields like healthcare, education, and public administration, what practical auditing mechanisms should institutions prioritize right now?*

Prof. Ben Shneiderman: *Just as there are many layers of governance for planes, housing, and medical care, there will be many layers of governance for AI based systems: guidelines for developers, guardrails for systems, auditors to review, insurance companies to compensate for damages, and government agencies to protect citizens. I favor incident reporting systems that enable professionals and the general public to describe the harms they have seen or suffered. Transparency and openness are helpful, but knowledgeable watch dogs and journalists are needed to raise alarms and push for safer systems. I believe strongly in more than responsible design, I speak for the importance of clarity about responsibility, legally and morally. Only people and their organizations are responsible, so that has to guide designers in creating audit trails or logs to enable discovery when failures occur and to understand how clever people have prevented failures.*

4. About User Interfaces and Explainability

Antonio Paulo: *You have long demonstrated that direct manipulation and rich information visualization provide superior user understanding compared to natural language black-boxes. How should the next generation of software interfaces evolve to give users meaningful control, auditability, and clear mental models over complex algorithmic outputs?*

Prof. Ben Shneiderman: *I believe that a blend of natural language windows combined with control panels will give users what they need to more clearly specify their intents. While the shopping interface for Amazon has its flaws, it is an excellent example of designs which build user self-efficacy and clarify their responsibility, while providing some mechanisms for conflict resolution. Their user interfaces rely on check boxes, sliders, menus, and form fill-in designs that enable rapid exploration of alternatives and easy reversal of actions.*

5. A Message to Future Practitioners

Antonio Paulo: *What core advice would you offer to software developers, designers, and interdisciplinary students who want to build technologies that genuinely put human values, rights, and well-being at the center of innovation?*

Prof. Ben Shneiderman: *The basics of good interfaces are that they are comprehensible, predictable, and controllable. The users must comprehend what is happening, be able to predict what the outcome of a request will be, and have meaningful control of the system to get what they want, as well as to easily revise a request to get what they want. Like successful bank machines, the users should know how to walk up and get \$60. Users rejected chatty bank machines that invited an open-ended conversation with "How can I help you?", preferring the clear, but limited functions, that users most wanted. Modern supertools use AI services, and provide a much higher level of automation, while ensuring human control. Amplifying, augmenting, empowering, and enhancing human performance are the proper goals. Advanced AI-based systems are startlingly impressive, so people should learn to use them, but they are alarmingly flawed, so users should be careful, because they are responsible for their actions. Machines that are comprehensible, predictable, and controllable give people superpowers, helping them balance the powerful forces in companies and governments. Human dignity, social justice, and greater equality are grand challenges, but these aspirations will guide us to designing effective supertools.*